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3 **0. Introduction**

4
5 **1. Pre-requisites for Full Closed Loop**

- 6 1.1 Well tuned hybrid closed loop
- 7 1.2 Fast insulin
- 8 1.3 Reliable insulin delivery from pump and cannula
- 9 1.4 Excellent CGM
- 10 1.5 Meal-related limitations?
- 11 1.6 Lifestyle-related limitations?
- 12 1.7 Time required for setting-up

13 [Case study 1.1: Occlusion](#)

14 [Case study 1.2: Comparing insulins for FCL](#)

15 [Case study 1.3: Jumpy CGM](#)

16 [Case study 1.4: Lost pump connection](#)

17 [Case study 1.5: Permanent CGM values w/ 2x G6](#)

18 [Case study 1.6: When catastrophe strikes...](#)

19
20 **2. General Settings for Full Closed Loop**

- 21 2.1 SMB range extension
- 22 2.2 Max and min autoISF ratio
- 23 2.3 SMB delivery ratio
- 24 2.4 iobTH (iob_threshold_percent)
- 25 2.5 Eating Soon TT?
- 26 2.6 General settings in AAPS/Preferences

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28 **3. Description of autoISF / guidance by developers**

- 29 3.1 Overview
- 30 3.2 ISF modulation flowcharts
- 31 3.3 Exercise mode and dynamic iobTH
- 32 3.4 Automation options with autoISF parameters
- 33 3.5 Activity monitor
- 34 3.6 Using one-minute CGM (Libre 3)
- 35 3.7 AutoISF parameters overview table
- 36 3.8 Emulator for logfile analysis and tuning
- 37 3.9 Links to related case studies/detailed doc.s

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39 **4. Meals: Setting ISF_weights in AAPS/Preferences**

- 40 4.1 Getting started
- 41 4.2 bgAccel_ISF_weight
- 42 4.3 pp_ISF_weight
- 43 4.4 bgBrake_ISF_weight
- 44 4.5 dura_ISF and bg_ISF
- 45 4.6 Tuning your initial settings
- 46 4.7 Complex scenarios
- 47 4.8 Profile helper

48 [Case study 4.1: Pizza](#)

49 [Case study 4.2: Low carb meals](#)

50 [Case study 4.3: Hands-off FCL around Christmas](#)

51
52 **5. Temp. modulation of autoISF aggressiveness**

53 **5.1 Automatic modulation of loop aggressiveness**

- 54 5.1.1 autoISF off outside of meal windows
- 55 5.1.2 SMB off @ odd profile target
- 56 5.1.3 SMB off @ odd temp. target
- 57 5.1.4 Automatic diff. of FCL aggressiveness via Automations

58 5.1.5 Automatic diff. of FCL aggressiveness via Activity Monitor

59 5.1.6 Pro/con completely hands-off FCL

60 5.2 Manual modulation of FCL aggressiveness via DIY cockpit

61 5.2.1 Status recognition

62 5.2.2 Manual interventions from DIY cockpit

63 5.2.2.1 Temp. %profile or TT settings

64 5.2.2.2 Temp. settings in /preferences

65 5.2.2.3 Grey DIY cockpit buttons for pre-programmed FCL responses

66 5.2.3 Temporary exit from FCL

67 5.3 Recognizing loop state from the AAPS home screen

68 5.3.1 Modulated loop aggressiveness (3 top buttons)

69 5.3.2 Color scheme of the top 3 buttons

70 5.3.3 Active Automations

71 5.3.4 FCL related indicator fields

72 5.3.5 Overall **AAPS home screen**

73 5.3.6 Info given every 5 minutes in the **SMB tab**

74 5.3.7 Info about last 15 autoISF decisions

75 5.3.8 **SMB tab info when operating 1-minute Libre**

76 5.3.9 Summary: Your personal FCL cockpit

77 5.4 Ideas for an improved cockpit

78 5.4.1 Violet FCL icon and underlying buttons

79 5.4.2 Bottom buttons "insulin" etc.

80 5.4.3 Top three fields

81 5.4.3.1 TT dialogue field

82 5.4.3.2 Exercise button / dialogue field

83 5.4.3.3 Profile dialogue field

Skip what is in green writing:

= Drafted fragments or
not implemented ideas.

(Might be weeded out soon, many
things not really needed)

84 Case study 5.2: Sweet snacks / Glühwein w/ DIY cockpit

85 Case study 5.3: Compression low

86 6. Temp. modulation for exercise and light (In-)activity

87 6.1 Dynamic iobTH and sensitivity ratio

88 6.1.1 Manual (direct) iobTH modulation

89 6.1.2 Automations for iobTH modulation

90 6.1.3 Dynamic iobTH

91 6.2 Temp. % profile switch

92 6.3 DIY cockpit based on User action Automations

93 6.4 Improved FCL cockpit

94 6.4.1 Manual (direct) iobTH modulation

95 6.4.2 pre-set 4 kinds of exercise

96 6.4.3 optional meal pre-sets

97 6.4.4 optional hypo management pre-sets

Skip what is in green writing:

= Drafted fragments or
not implemented ideas.

(Might be weeded out soon, many things
not really needed)

98 6.5 Mastering the exercise after meal challenge

99 6.5.1 Manual mode

100 6.5.2 DIY cockpit button for User action Automation

101 6.5.3 Using pre-sets in improved FCL cockpit

102 6.6 Activity monitor based on step-counter

103 Case study 6.2 Biking day with hi carb lunch; DIY cockpit

104 7. Advanced HCL (meal announcement via pre-bolus)

105 7.1 Hurdles for FCL

106 7.2 Getting ready to advance from HCL

107 7.3 Reduced pre-bolus

108 7.4 Tuning autoISF in HCL

109 7.5 Dealing with disturbances/ins. sens/resistance

110 7.6 Exercise management

111 7.7 Remote control (small children) (fragment, to be completed /NN)

112 7.8 Other methods w/ meal announcement (MA)

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137	Case study 13.2: Using dynISF for FCL (sorry, nobody provided interpretable data yet)
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